Building Diagnostic Robotics

Commercial Roofing

Roof Quantity Take-Off

Project	Roof System	Assembly	Date
Sample (S2)	**60-mil TPO**	**4 in Polyiso - R-24**	**July 8, 2026**

Field Area
2,511 SF
25.1 squares

Roof Perimeter
237 LF
Coping and base flashing run

Roof Assembly
TPO / 4 in ISO
R-24 - 2 layers - plywood deck

Roof Geometry

Front / bottom width	**29 ft 5 in**	29.42 ft
Left side	**74 ft 6 in**	74.50 ft
Right side	**96 ft 3 in**	96.25 ft
Sloped back edge	**36 ft 7 in**	36.58 ft
Top-right clip	**5 ft 6 in**	5.50 ft
Parapet height	**4 ft 0 in**	4.00 ft
Base flashing height	**1 ft 6 in**	18 in

Modeled as right trapezoid: Field = W × (Lleft + Lright) ÷ 2. Dimensions per sketch "Midland (S2)."

Roof Assembly

Membrane	60-mil TPO - mechanically attached or fully adhered (see options)
Insulation	Two layers of 2 in polyiso (4 in total), joints staggered - R-24
Cover board	Optional 1/2 in HD-ISO / gypsum (recommended, not in totals)
Deck	Existing plywood
Parapets	4 ft high; TPO base flashing 18 in up wall; metal coping at top
Perimeter	Metal cap flashing (coping) over continuous termination bar

Quantity Take-Off

Item	Basis / Calculation	Qty	Unit	Notes
Membrane - 60-mil TPO				
TPO membrane - field	Field area x 1.10 waste	**2,763**	SF	Heat-welded seams
TPO base flashing - parapets	237 LF x 1.5 ft x 1.10	**391**	SF	18 in up wall; adhered
TPO rolls (10 ft x 100 ft)	(Field + flashing) / 1,000 SF	**4**	rolls	Field + base flashing
TPO cover strip	Detailing / seam allowance	**1**	roll	T-joints, patches
Insulation - Polyisocyanurate				
Polyiso - 2 layers (4 in total)	Field x 2 layers x 1.07	**5,374**	SF	R-24 LTTR; stagger joints
Polyiso boards (4 ft x 8 ft)	5,374 SF / 32 SF per board	**168**	boards	84 boards per layer
Insulation fasteners + plates	Approximately 5 per board, both layers	**785**	EA	Enhance perimeter / corners

Item	Basis / Calculation	Qty	Unit	Notes
Edge Metal and Termination				
Metal cap flashing / coping (10 ft)	Perimeter 237 LF x 1.05	**25**	pcs	Approximately 249 LF; pre-finished
Coping splice / cover plates	1 per joint	**24**	EA	Concealed splices
Coping corners (in / out)	Field count	**5**	EA	Prefab or fabricated
Termination bar (10 ft)	Perimeter 237 LF x 1.05	**25**	pcs	Top of base flashing
Drains and Penetrations				
Roof drain assemblies - “A”	Marker “A” count	**2**	EA	Clamp ring + strainer
Tapered sump kits	1 per drain	**2**	EA	Recommended - flat ISO
Curb / penetration flashing - “B”	Marker “B” - 39 in x 19 in	**1**	lot	TPO wrap + termination bar
Sundries and Accessories (Allowances)				
Bonding adhesive - base flashing	355 SF / 60 SF per gal	**6**	gal	Both attachment options
Pre-molded corners (in / out)	Detailing allowance	**10**	EA	Drains, curb, parapets
Water cut-off mastic	Detailing allowance	**3**	tubes	Drains, terminations
Cut-edge sealant (TPO)	Detailing allowance	**3**	bottles	Exposed cut edges

> **Optional - Cover Board (not included in totals):** 1/2 in HD-ISO or gypsum cover board over the polyiso - approximately 83 boards (4 ft x 8 ft, field x 1.05). Recommended for improved puncture resistance, wind uplift, and warranty terms. Add if specified.

Membrane Attachment - Select One

Option A - Mechanically Attached — Lower Cost

Membrane fasteners + plates	**252 EA**
Bonding adhesive (base flashing)	**6 gal**

Fasteners in the seams, with tighter patterns at perimeter and corner zones. Faster installation and lower material cost; wind performance is governed by the fastening pattern.

Option B - Fully Adhered — Best Finish

Field bonding adhesive	**42 gal**
Bonding adhesive (base flashing)	**6 gal**

Membrane adhered across the field (approximately 60 SF per gal, both surfaces). Smoother finish and strong wind performance; higher adhesive cost and temperature-sensitive installation.

Assumptions and Notes

Waste factors: 10% membrane, 7% insulation, 5% metal. Adjust for crew, roof complexity, and off-cuts.

Geometry: Field area modeled as a right trapezoid from the sketch. The 5 ft 6 in top-right clip removes less than 1% (approximately 15 SF) and is absorbed by waste; field-verify dimensions before ordering.

Wind and fastening: Fastener counts are baseline. Final pattern per ASCE 7 wind zone and FM / manufacturer requirements; enhance perimeter and corner zones.

Drainage: Two tapered sump kits are included since the polyiso is flat (no positive slope). Confirm tapered-ISO / slope scope separately if required.

Cover board: Excluded from totals; recommended for durability and warranty (approximately 83 boards if added).

Deck: Existing plywood assumed sound and fastenable; verify condition and fastener pull-out values.

Scope: Material planning take-off. Confirm all quantities against final approved drawings and the selected manufacturer’s system requirements.